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REGULAR PAPER

Dimension Reduction of Point Cloud-Based Obstacle Representation via Potential Fields for Whole-Body Collision-Free RL Manipulation

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Abstract

Reinforcement learning (RL) is effective in complex manipulation tasks; however, training RL policies in obstacle-rich environments remains challenging. Most existing RL models require very deep networks to encode obstacle information from high-dimensional point cloud data and rely on sparse collision penalties, resulting in inefficient learning. To address this limitation, this study proposes an obstacle-aware RL framework that introduces additional states and rewards derived from potential field-based repulsive forces. Because the proposed representation encodes the geometric relationships between the robot and obstacles using a fixed number of state variables, regardless of the number or shape of obstacles, it mitigates the learning instability caused by high-dimensional point cloud inputs. Experimental results demonstrate that the proposed representation is sensitive to the relative spatial relationship between robots and obstacles. Furthermore, the proposed model achieved a success rate of over 81.1% across various unseen environments, demonstrating strong generalization performance. In complex tasks such as opening a door with and without a lever, the proposed method achieved success rates of 65.0% and 75.3%, respectively, without collisions. These results demonstrate the effectiveness of repulsive force-based states and rewards in enabling collision-free RL manipulation in complex scenarios.

Keywords Reinforcement learning · Potential field · Obstacle representation · Point cloud · Signed distance function · Whole-body collision-free control

1 Introduction

Manipulators are increasingly deployed in diverse applications, including manufacturing [1–4], logistics [5, 6], and households [7–9]. Despite recent advances in learning-based control, such as imitation learning and reinforcement learning (RL), the risk of collisions remains high in obstacle-rich environments. Furthermore, with the growing interest in human–robot collaboration, safety and collision avoidance have become increasingly critical. For industrial tasks involving the interiors of large components, collision avoidance must encompass not only the end effector but also the entire body of the robot.

Collision-free motion can be planned by incorporating collision constraints into sampling- [10] or optimization-based motion planners [11, 12]. However, these traditional planners are primarily limited to simple tasks, such as moving end effectors to a target position and orientation. They face limitations in performing complex and continuous tasks owing to high computational costs and challenges associated with real-time replanning. Learning-based motion policies have been widely used to overcome these limitations. Nevertheless, most existing studies have validated the collision avoidance capability solely for simple reach tasks [4, 13, 14], thereby limiting their applicability. Executing high-level industrial tasks, such as assembly, manipulation, and collaborative work, requires learning motion policies that can perform complex tasks without collisions. Therefore, research on learning-based collision avoidance motion

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policies is essential for ensuring safety during the execution of complex tasks.

Appropriate state information is crucial, particularly in obstacle-rich environments, for training a policy to successfully learn and execute tasks. In learning-based motion policies, obstacles are commonly represented by approximating them as primitive geometries [15–17], such as spheres and cylinders, or by utilizing entire point clouds [17, 18]. Primitive geometry-based representations typically rely on specific feature points, such as centers, to define obstacle locations. For example, human arms are represented by cylinders to calculate the distances from the arm to a robot [17]. However, because the number of inputs in an RL model is typically fixed, these approaches are limited to environments with a fixed number of obstacles. Furthermore, primitive geometries lack the precision required to capture complex and irregular geometric shapes, which are frequently observed in the real world.

Point cloud representation can mitigate the limitations of primitive geometry-based representations. This approach employs a fixed input size using a latent vector generated by embedding a voxelized point cloud through an encoder [19]. Similarly, imitation learning policies can be trained by embedding point clouds using PointNet++ [18]. Although existing point cloud-based approaches provide rich information on obstacles, they must often cope with excessively high-dimensional state-space data. This can overwhelm the task-relevant state information required for motion generation, thereby hindering the stable convergence of policies that must simultaneously learn both manipulation and collision avoidance. End-to-end training on a deep encoder and policy network can be unstable because RL models tend to lose plasticity in deep networks [20]. Although the spatial relationship between robots and obstacles is the most important information for planning collision-free motion, existing methods fail to accurately represent this information. Therefore, essential and compact information on obstacles is required to effectively train RL agents.

The robot geometry should also be considered to achieve whole-body collision-free control. Primitive geometries and signed distance functions (SDFs) have been widely used to represent robot geometries. For instance, a sphere representation can be used to model a robot shape and evaluate the likelihood of collision [21]. Although this approach requires simple computations, it poses a scalability challenge. When many spheres are used to precisely capture the robot geometry, the computational cost increases significantly. Alternatively, an individual robot link can be treated as a cylinder, to calculate the distances for RL inputs [15]. However, these methodologies remain limited in their ability to precisely represent the complex shapes of robots. By contrast, SDFs represent a robot geometry by returning the minimum distance from any point in space to the robot surface. Because SDFs explicitly represent distance, they are spatially differentiable. Therefore, SDFs have been widely adopted in model-based collision avoidance motion planners [22–24]. However, the real-time computation and deployment issues remain unresolved. This problem has been addressed using a neural robot SDF model trained to predict the distances and their gradients at the current poses of robots [25]. Furthermore, a whole-body SDF was developed by representing the SDF of an individual link and utilizing it to execute forward kinematics [26].

This study proposes a new collision avoidance motion policy that uses repulsive forces generated by artificial potential fields (APFs) [27] as the states of RL to address the limitations of existing learning-based motion policies for collision avoidance and task scalability. First, in contrast to previous studies that rely on centroid-based distances between the robot and obstacles, the proposed method considers the distance between the robot surface and obstacle surfaces. This approach enables the policy to effectively distinguish differences in object shape and size, even when the centroids of the obstacles are identical. Second, point cloud data are not directly used as inputs to the RL agent. Instead, a repulsive force vector for each robot link represents the spatial relationship between obstacles and links. Furthermore, rather than directly using the gradient of the potential field as the repulsive force, it is modified to facilitate the RL training process. This approach significantly reduces the number of layers and parameters in the network, thereby improving training stability and collision-free motion planning. Furthermore, this study demonstrates that this low-dimensional representation can facilitate training for complex manipulation tasks.

The remainder of this paper is organized as follows. Section 2.1 describes the overall framework and problem formulation, Section 2.2 explains the formulation of repulsive forces based on obstacle state information, Section 2.3 introduces the robot SDF used to compute the repulsive forces, and Section 2.4 details the reward design and shaping strategy. Section 3 presents experimental evaluations of the proposed approach. Finally, Section 4 concludes the paper and discusses the limitations of this study and directions for future research.

2 Method

This study proposes a repulsive force-based obstacle-aware representation for the state in RL-based manipulator control. This method converts the observed point cloud into repulsive forces and transforms them into a lower-dimensional representation that can be utilized as input for the policy.

This section describes the methodology used to compute and apply these forces.

Since RL models determine their actions and estimate value functions using states, they should represent robot motion, task objectives, and environments. The state vector comprises 1) goal-conditioned information for task execution (e.g., the opening angle in a door-opening task), 2) a repulsive force-based obstacle representation, and 3) proprioceptive information, such as the joint angles of the robot, as shown in Fig. 1.

2.1 Repulsive Force-Based Obstacle-Aware Representation

Among the various APFs, an exponential APF was introduced because its gradient is smooth, even when obstacles are close to the robot. Because this study focuses on obstacle-rich environments, a spatially continuous and smooth function is desirable for stable RL training. The gradient of the exponential APF can be calculated as $\nabla U = \sum_j \exp(-\alpha d_{ij}) \nabla d_{ij}$, where α is a coefficient and d_{ij} is the distance between the ith link and a point j on an obstacle surface. Here, the summation is required because the spatial relation between the link and several points on the obstacles should be considered. If this gradient ∇U is directly used as a repulsive force, it can misrepresent the collision risk. For example, when a link is surrounded by obstacles from several directions, the individual repulsive force vectors cancel each other out. Consequently, ∇U can be very small even when the collision risk is high. To address this issue, this study proposes a new repulsive force. In particular, the direction of the repulsive force is determined by summing $\exp(-\alpha d_{ij}) \nabla d_{ij}$. Its magnitude is determined by the closest obstacle point. The repulsive force $\boldsymbol{F}_{rep,i}$ of the ith link is defined as:

$$\boldsymbol{F}_{\mathrm{rep},i} = \frac{\sum_j \exp(-\alpha d_{ij}) \nabla d_{ij}}{\left\| \sum_j \exp(-\alpha d_{ij}) \nabla d_{ij} \right\|} \exp\left[-\alpha \min_j (d_{ij}) \right] \tag{1}$$

This approach ensures semantically consistent information, regardless of the obstacle size or shape. Repulsive forces are generated for each link of the manipulator. Note that a repulsive force is not computed for the two links close to the base because they are unlikely to collide with obstacles. Figure 2 illustrates the link-wise repulsive forces generated from the obstacle point cloud, where each arrow represents the repulsive force vector acting on the corresponding robot link.

2.2 Whole-body SDF

The robot SDF is required because both the distance d_{ij} and its gradient ∇d_{ij} are necessary to calculate the repulsive force, as expressed in Eq. 1. For effective whole-body collision-free control, the individual SDF for each link of the robot must be calculated [26]. Because a manipulator comprises multiple links, the geometry of the entire robot changes according to the joint configuration. If a whole-body SDF is defined directly in the global frame, it becomes inherently pose-dependent and requires recalculation for every possible joint configuration. To avoid this issue, pose-invariant SDFs are obtained for individual links in their respective body frames. These link-level SDFs are then composed using forward kinematics according to the current joint configuration, enabling efficient whole-body distance computation without recomputing a global SDF.

To obtain the SDF, this study adopted a neural SDF approach [28] in which the SDF was approximated using a neural network. The network architecture is illustrated in Fig. 3. The output of the SDF is the distance d_{ij}. The input of the neural SDF comprises an embedding vector representing the link index and the position of point j, where

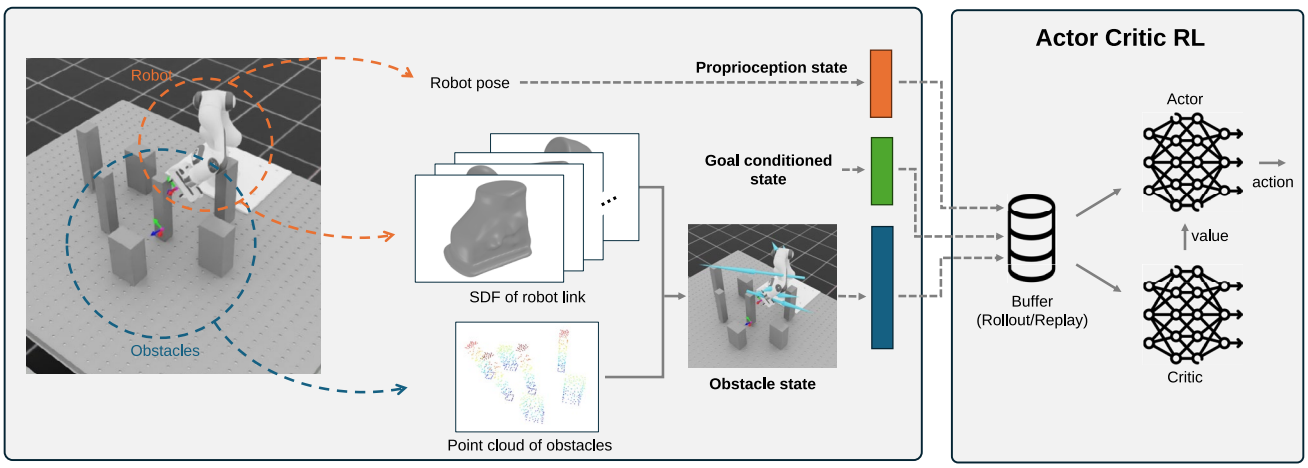


Fig. 1 Overview of the proposed model

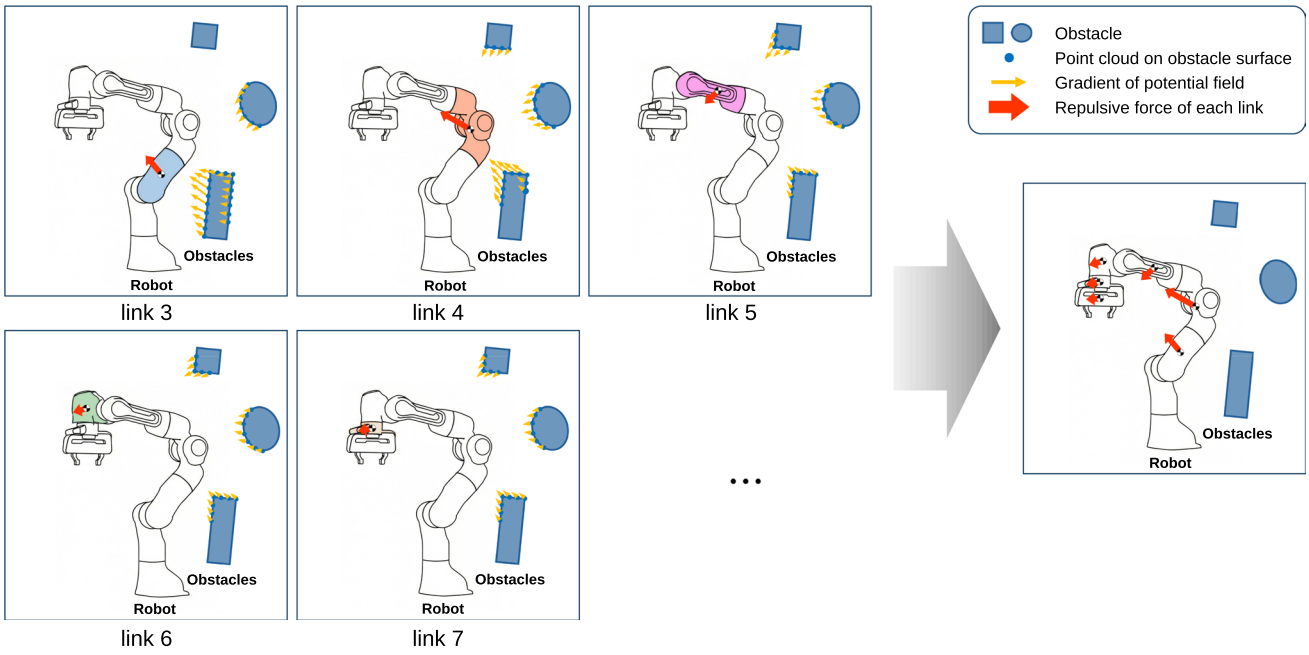


Fig. 2 Link-wise repulsive forces generated by the obstacle point cloud

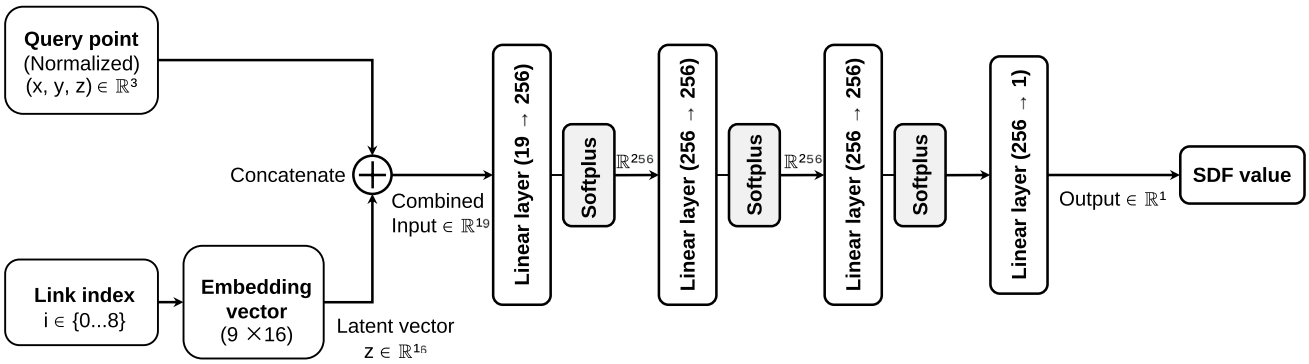


Fig. 3 Architecture of the neural SDF network. The network receives a query point and an embedded link index as inputs and consists of three hidden layers, each with 256 hidden units

the position (x, y, z) is represented in the link body frame. Notably, a single neural network is trained to approximate the SDFs of all links simultaneously to reduce inference time. The network consists of a three-layer perceptron, as illustrated in Fig. 3. Softplus [29] was used as the activation function because its gradient is more continuous than that of ReLU.

A dataset for training the neural SDF was generated based on a mesh model of robot links. One hundred thousand points were sampled from each link to train the neural SDF. According to the eikonal equation, the magnitude of the gradient should be 1 for the SDF. Considering that the SDF is a spatial function that returns the minimum distance from any point in space to a given surface, its output increases at a unit rate when moving away from the surface along the normal direction. Thus, the loss function of the neural SDF can be defined as the sum of the mean squared error L_{mse} and eikonal regularization loss $L_{eikonal}$ [30], as follows:

$$\begin{aligned} &\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{eikonal}}, \\ \text{where} \quad &\mathcal{L}_{\text{mse}} = \frac{1}{N}\sum_{j=1}^{N} |\text{SDF}(\boldsymbol{X}_j, \boldsymbol{E}_i) - \tilde{d}_{ij}|, \\ &\mathcal{L}_{\text{eikonal}} = \frac{1}{N}\sum_{j=1}^{N} (\|\nabla_{\boldsymbol{X}_j} \text{SDF}(\boldsymbol{X}_j, \boldsymbol{E}_i)\| - 1)^2. \end{aligned} \tag{2}$$

where $\boldsymbol{X}_j$ denotes the three-dimensional coordinates of point j, $\boldsymbol{E}_i$ is the embedding vector of the ith link, and $\tilde{d}_{ij}$ is the true distance between point j and the surface of the ith link. The network was trained to predict the SDF with an average accuracy of 0.00342. The neural SDF accurately predicted the signed distance, as shown in Fig. 4.

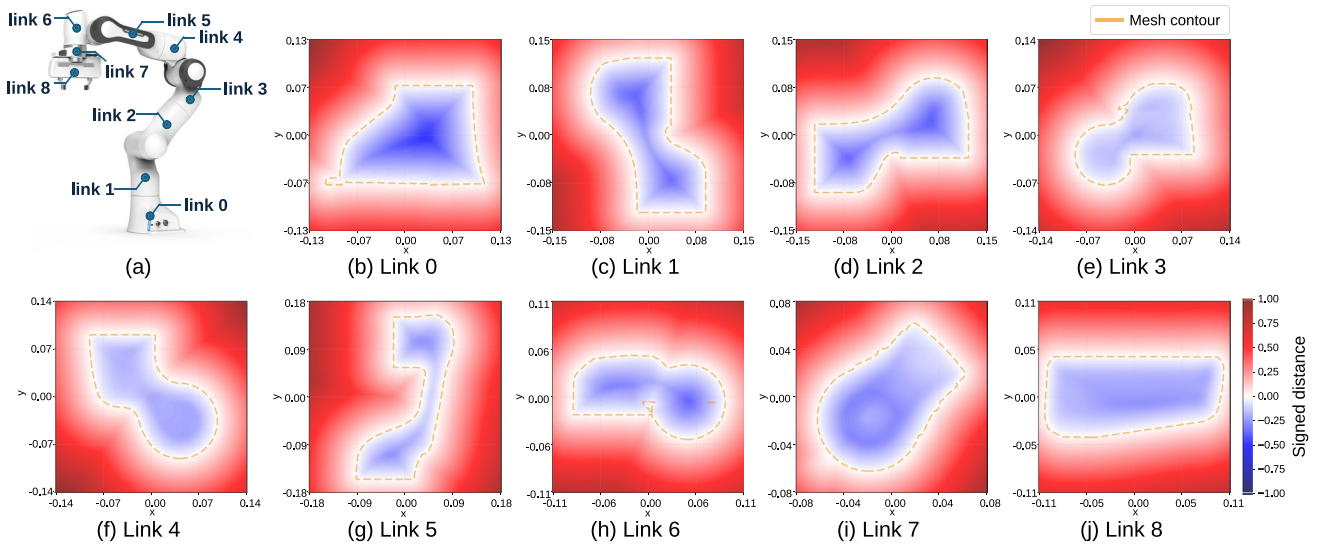


Fig. 4 Predictions of the neural SDF. (**a**) Configuration of the Franka manipulator; (**b**)–(**j**) Estimated SDF for each link. The yellow dotted line indicates the surface of the links

2.3 Collision-related Reward

A dense reward for collision can result in stable training and fast convergence for learning collision-free motion. In previous RL models [2, 31, 32], a sparse collision reward was used by simply providing a constant penalty value when collisions occurred. Although this sparse reward function is easy to use, RL models require more samples and iterations to learn the collision avoidance motion. Thus, the latent vectors obtained from the point cloud have been typically used for dense rewards. However, this approach is challenging because the mapping from the point cloud to the latent vector remains ambiguous.

The repulsive force proposed in this study can effectively provide dense rewards for collision avoidance. Because the repulsive force expressed in Eq. 1 explicitly includes the spatial relations between the robot links and obstacles, it can be directly used to design a dense reward. Moreover, link-wise collision penalties can be beneficial when whole-body collision-free control is considered. Because the proposed model calculates the repulsive force of each link, all the forces can be included in the repulsive force-related reward as follows:

$$r_{\text{repulsive}} = \sum_i \exp[-\beta \min_j(d_{ij})], \tag{3}$$

where β is a parameter for adjusting the effects of repulsive forces on the RL training. Other reward terms associated with a given task can be added to $r_{repulsive}$ to accomplish the task while preventing collisions.

2.4 Training

To use the proposed model, the neural SDF should be trained first, followed by RL training. If an end-to-end training approach is adopted, the network becomes excessively deep, leading to the loss of RL plasticity. By using the pre-trained neural SDF in the RL training process, the RL policy can focus on collision avoidance and task execution because the neural SDF has already encoded the spatial relationship between the obstacle and robot links. The repulsive force-based obstacle representation explicitly provides distance and directional information regarding obstacles. In addition, a continuous dense reward can be effectively formulated based on distance information. This provides explicit obstacle-avoidance signals before collisions occur. By using the repulsive force in the state and reward, the proposed model can facilitate the efficient and stable training of collision-free motion policies.

The training was conducted using Robosuite [33] and IsaacLab [34]. A soft actor critic [35] was used to conduct various complex tasks with a long horizon to mitigate catastrophic forgetting and optimize efficiency.

3 Results and Discussion

This study validated the consistent mapping of robot-obstacle spatial relationships (Section 3.1), the effects of the repulsive force on task completion (Section 3.2), and the applicability of the representation to a multi-stage task (Section 3.3).

3.1 Spatial Relationship Consistency

To verify that the repulsive force consistently represents the spatial relationship between the robot and obstacles, the force vectors were observed when the shapes, sizes, and positions of the obstacles varied. In particular, the RL model was trained to reach the target position and orientation in an environment in which the obstacle positions, sizes, and shapes were fixed. This environment is referred to as the reference environment, as shown in Fig. 5(a). The model was trained using reward terms outlined in Table 1. Subsequently, the robot action sequence in this fixed environment was applied in different environments where the shapes, sizes, numbers, and positions of the obstacles were different, as shown in Fig. 5(b)–(d). The obstacles used included rectangular boxes, mustard bottles, and water pitchers. The obstacle was either the original size or twice the original size. Subsequently, the repulsive force vector in the reference environment and the vectors in other environments at the same time step were determined. The L2-norm of the difference between these two vectors was then calculated to verify the similarity between the two repulsive vectors.

When the obstacle position changed, the L2-norm error was large; however, the obstacle size and shape remained the same, as shown in Fig. 6(a). Because positional changes significantly affect the spatial relationship between the robot and obstacle, this large L2-norm error indicates that the repulsive force successfully represents this spatial relationship. By contrast, the L2-norm shown in Fig. 6(b) is small because the effects of the obstacle shape on the spatial relationship were small. The L2-norm was also small when the obstacle size changed, as shown in Fig. 6(c). Note that the shape variation was likely to change the direction of the repulsive force, whereas its magnitude was less affected. The size variation tended to change the magnitude of the repulsive force, whereas the direction changed only slightly. Consequently, the L2-norm values associated with the shape and size variations were similar.

Also, as shown in Fig. 6(b) and (c), the distribution of the L2-norm related to shape changes was wider than that

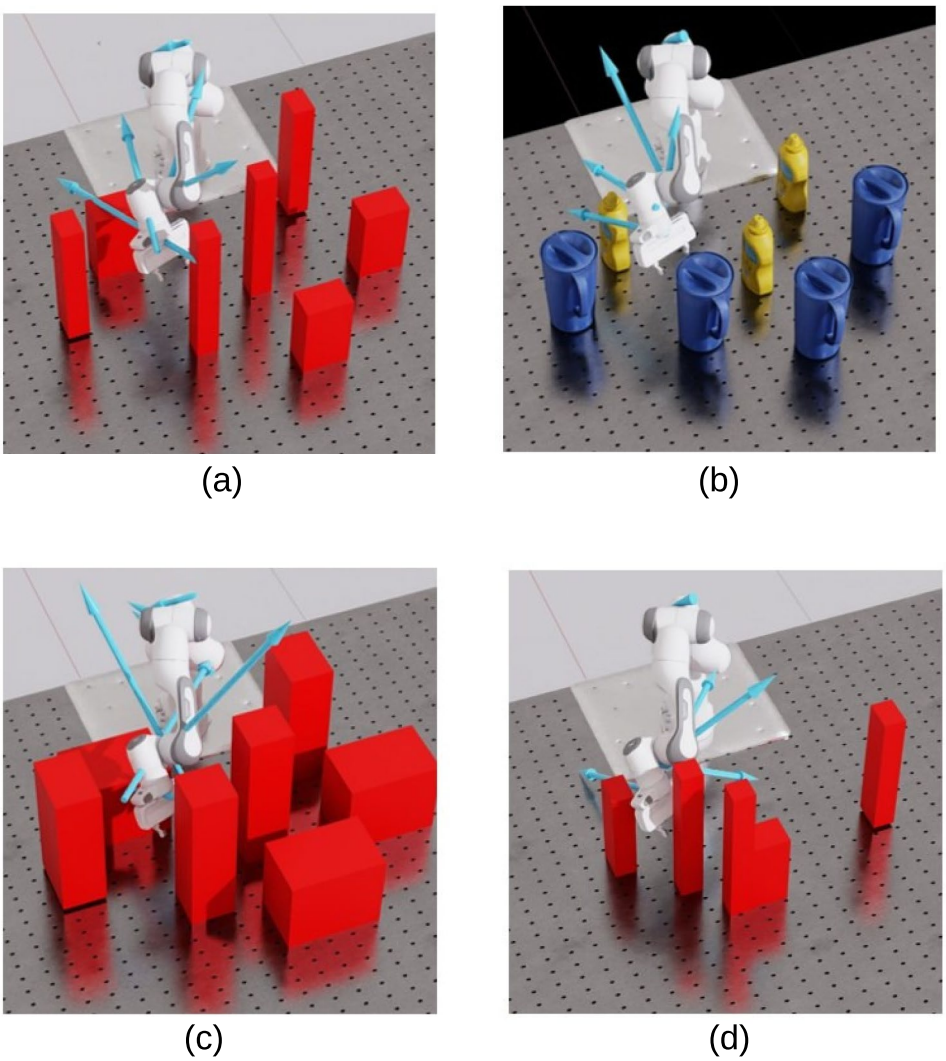


Fig. 5 Examples of obstacle-rich environments. The repulsive force of each link is represented by arrows: (**a**) reference environment; (**b**) environment where the obstacle shapes are changed compared to those in the reference environment, while the positions and sizes remain unchanged; (**c**) environment where the obstacle sizes are changed compared to those in the reference environment, while the positions and shapes remain the same; (**d**) environment where the obstacle positions and numbers are changed compared to those in the reference environment, while the sizes and shapes remain the same

Table 1 Reward terms

Goal	
Relative position reward	$1 - \tanh\left(\frac{\|p_{\text{goal}} - p_{\text{ee}}\|_2}{0.3}\right)$
Relative position	$1 - \tanh\left(\frac{\|p_{\text{goal}} - p_{\text{ee}}\|_2}{0.1}\right)$
Relative orientation reward	$1 - \tanh\left(\frac{\Delta q}{0.3}\right)$
Collision	
Repulsive force penalty	$\sum_i \exp\left[-\gamma \min_j(d_{ij})\right]$
Collision event	$\begin{cases} 1, & \text{if collision occurs} \\ 0, & \text{otherwise} \end{cases}$
Success	
Success reward	$\begin{cases} 1, & \text{if } \|p_{\text{goal}} - p_{\text{ee}}\|_2 < 0.3 \\ & \text{and } \Delta q < 0.3 \\ 0, & \text{otherwise} \end{cases}$

where p_{goal} denotes the position of the goal point, p_{ee} denotes the position of the end effector of the manipulator, q_{ee} denotes the orientation of the end effector, Δq denotes the quaternion difference between the end effector and goal orientation, and γ denotes coefficient of repulsive force penalty

related to size changes. The difference between the surface of the box and the surface of the mustard bottle (or water pitcher) varied with height. For example, at a low height, the surface of the mustard bottle may be similar to that of the box, but at a higher height, the surface can be very different. These results suggest that the proposed repulsive force not only captures the positional relationship between the robot and the obstacles but also accounts for variations in their shape and size.

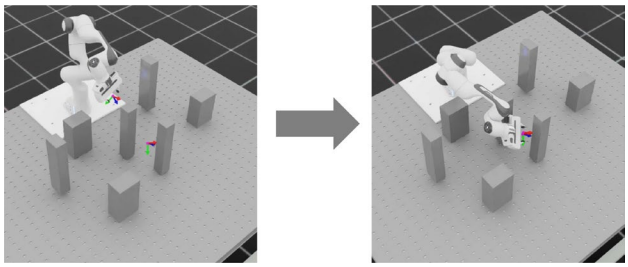


Fig. 7 Reach task used for comparison with a baseline model. (**a**) and (**b**) depict the initial and goal state of the robot, respectively

3.2 Ablation Study on Proposed Obstacle Representation

To validate the effects of the repulsive force on task completion, the success rates of the proposed model (i.e., RL with repulsive force) were compared with those of baselines: RL with centroid-based relative position, centroid-based repulsive force, PointNet, and PointNet++. In particular, the centroid-based relative position model and centroid-based repulsive force model can be classified as a primitive geometry-based approach. PointNet and PointNet++-based approaches used latent feature vectors composed of 64 and 128 features, respectively. All models were trained using the same reward function, hyperparameters, and dataset (Fig. 7).

The proposed model noticeably outperformed all baselines in every environment, as shown in Table 2. When the obstacle shape changed, the reduction in the success rate is relatively small for the primitive geometry approach compared to the encoder-based approach. This suggests that the encoder-based RL is highly sensitive to the obstacle

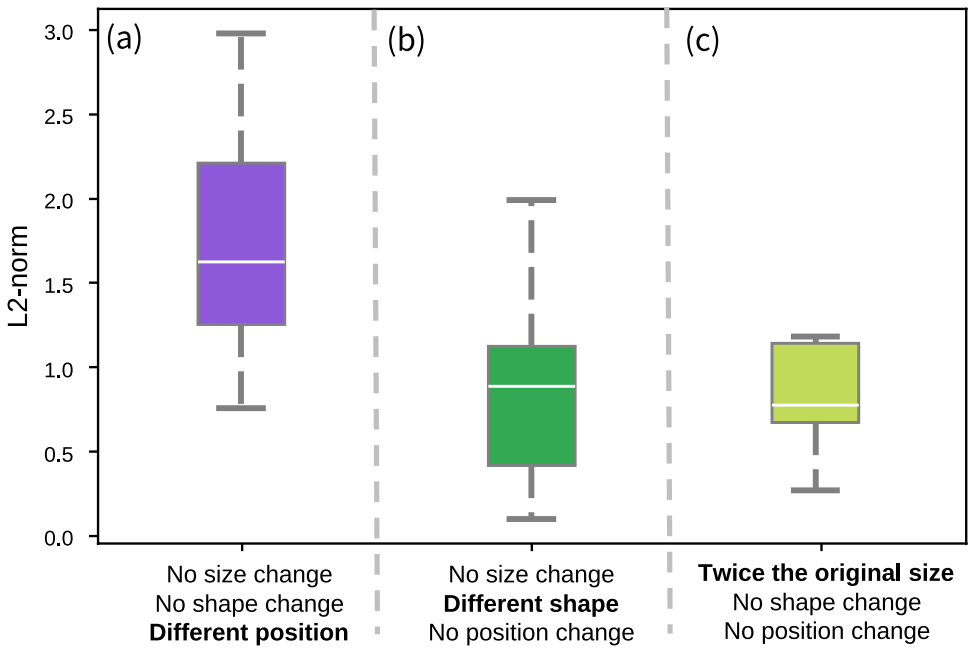


Fig. 6 Differences in the repulsive force due to obstacles. The box-and-whisker plots in (**a**) show the L2-norm of the repulsive forces between the reference environment and environment with difference sizes. (**b**) and (**d**) show the L2-norm when the obstacles have different shapes and positions, respectively

Table 2 Comparisons of success rates in multiple environments

Environments	Primitive geometry-based RL		Encoder-based RL		
	Centroid-based relative position	Centroid-based repulsive force	Point-Net based	Point-Net++ based	Proposed model
Reference environment [Fig. 5(a)]	70.5%	72.0%	77.0%	69.6%	**90.5%**
Environment with different obstacle shapes [Fig. 5(b)]	69.0%	67.0%	67.3%	60.5%	**85.5%**
Environment with different obstacle positions [Fig. 5(d)]	61.0%	50.0%	74.8%	58.0%	**79.0%**
Inference time (ms)	14.85	15.23	24.41	19.19	21.83

The bold entries are used to highlight the results of the proposed model

shape. However, the performance of the primitive geometry approaches significantly degraded when the obstacle position changed. The encoder-based approaches require longer inference times compared to the primitive geometry approaches.

3.3 Ablation Study on Collision-Related Reward

In this study, a collision-related reward based on the link–obstacle distance was introduced to effectively incorporate collision and distance information into the training process. To evaluate its effect, the distance between link surfaces and obstacle surfaces was compared with and without the collision-related reward, as shown in Fig. 8. The proposed reward maintained larger distances from the obstacles, with an average increase of 3.9 cm compared to the case without the reward. Consequently, the success rate improved by 8.5%, as provided in Table 3.

Table 3 Success rates with and without the collision-related reward

	w/o collision-related reward(Sec 3.3)	End Effector-only(Sec 3.4)	Proposed
Success rate	81.0%	30.5%	**90.5%**

The bold entries are used to highlight the results of the proposed model

3.4 Ablation Study on Link-Wise Repulsive Force

In this study, a link-wise repulsive force was used to achieve whole-body collision avoidance for the manipulator. To evaluate its effectiveness, the proposed method was compared to the case when the repulsive force was considered for the end-effector-only. For this end-effector-only repulsive force model, collision risks for other links are not considered during training. As a result, the policy failed to sufficiently explore the environment. The end-effector-only repulsive force model yields a success rate of only 30.5%, with 125 terminations out of 200 trials without task completion. In contrast, the link-wise repulsive force results in a 90.5% success rate with four terminations, as provided in Table 3.

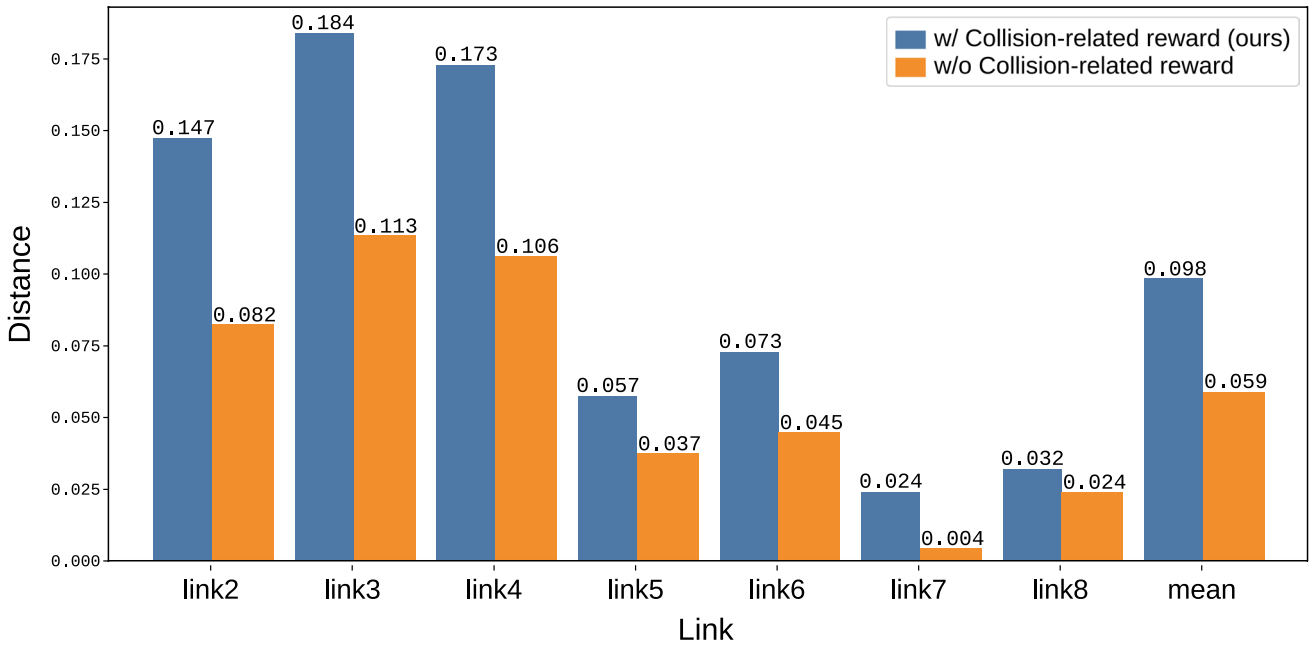


Fig. 8 Predictions of the neural SDF. (**a**) Configuration of the Franka manipulator; (**b**)–(**j**) Estimated SDF for each link. The yellow dotted line indicates the surface of the links

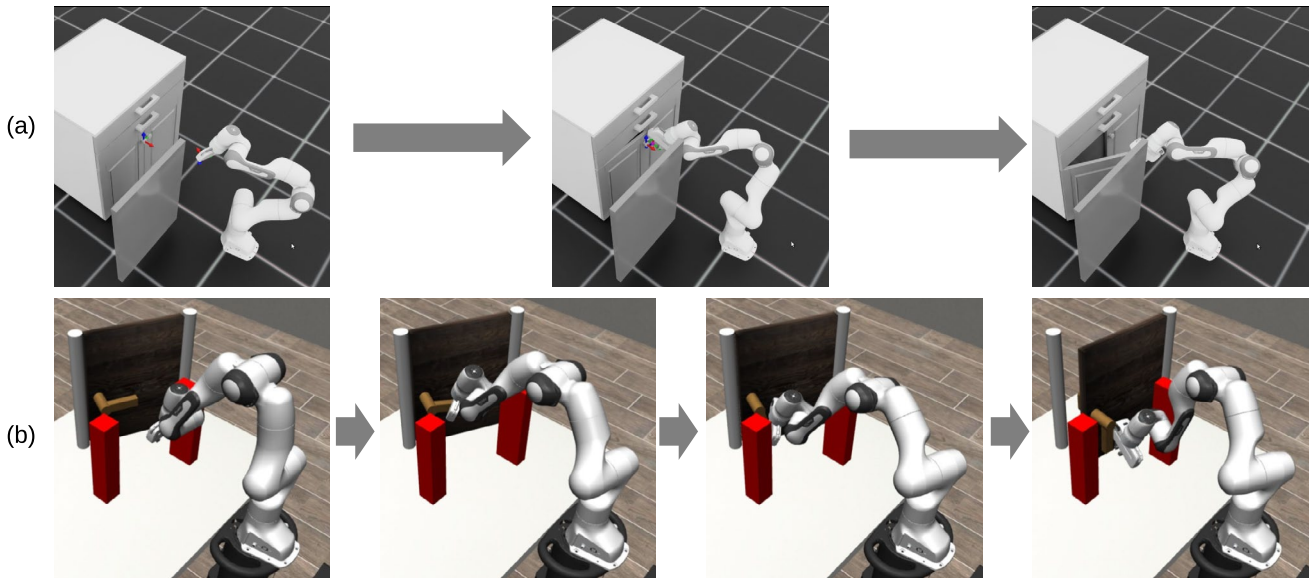


Fig. 9 Complex tasks with obstacles: (**a**) opening a door without a door lever; (**b**) opening a door with a door lever

3.5 Complex Task Execution Capability

The performance of the proposed model was validated for complex tasks, which are defined as multi-stage manipulation problems. They also involve several object interactions, kinematic constraints, and sequential decision-making, including opening a door without and with a door lever, as shown in Fig. 9. Note that because these tasks must be conducted within a highly constrained workspace, even a few obstacles significantly increase the task difficulty. Furthermore, for such complex operations, the position and size of obstacles are more critical than their absolute number. Therefore, one or two obstacles were placed at locations where interference with the task is significant.

Opening without a door lever can be achieved by sequentially conducting reach, grasp, and door-pull motions. Opening with a door lever requires a longer horizon task: reach, grasp, lever rotation, and door-pull motion. The RL model must execute this complex task while avoiding collisions, which is challenging. The details of the rewards for each task are presented in Table 4. Note that the collision-related reward expressed in Eq. 3 is independent of the task. Thus, the same reward term is used for all tasks in the event of a collision. Moreover, the reward terms associated with goal and success vary across the tasks.

The RL model successfully accomplished the given tasks without collisions. In particular, the robot opened a door (either the left or right door) without a door lever, with a success rate of 75.3%. The opening task with a door lever was accomplished with a success rate of 65.0%, which is high, considering the difficulty. These results demonstrate

Table 4 Reward terms

Opening a door without door lever	
Goal	
Relative position reward	$1 - \tanh\left(\frac{\|p_{\text{goal}} - p_{\text{ee}}\|_2}{0.5}\right)$
Relative orientation reward	$q_{\text{ee}} \cdot q_{\text{handle}} - 1$
Door open reward	$\begin{cases} \theta_{\text{door}}, & \text{if } \theta_{\text{door}} > 0.1 \text{ rad} \\ 0, & \text{otherwise} \end{cases}$
Collision	
Repulsive force penalty	$\sum_i \exp[-\gamma \min_j(d_{ij})]$
Collision event	$\begin{cases} 1, & \text{if collision occurs} \\ 0, & \text{otherwise} \end{cases}$
Success	
Success reward	$\begin{cases} 1, & \text{if } \theta_{\text{door}} > 0.4 \text{ rad} \\ 0, & \text{otherwise} \end{cases}$
Opening a door with door lever	
Goal	
Relative position reward	$1 - \tanh(10\|p_{\text{goal}} - p_{\text{ee}}\|_2)$
Lever rotate reward	$0.25 \frac{\theta_{\text{lever}}}{0.5\pi}$
Collision	
Repulsive force penalty	$\sum_i \exp[-\gamma \min_j(d_{ij})]$
Collision event	$\begin{cases} 1, & \text{if collision occurs} \\ 0, & \text{otherwise} \end{cases}$
Success	
Success reward	$\begin{cases} 1, & \text{if } \theta_{\text{door}} > 0.4 \text{ rad} \\ 0, & \text{otherwise} \end{cases}$

where p_{goal} denotes the position of the door handle or door lever; p_{ee} denotes the position of the end effector of the manipulator; q_{ee} and q_{handle} denote the orientations of the end effector and door handle, respectively; θ_{lever} denotes the rotation angles of the lever; θ_{door} denotes the door opened angle

that the proposed representation method enables stable learning, even for long-horizon tasks in which the number of stages increases. The values of the hyperparameters are provided in Appendix A.

3.6 Real-time Performance

To evaluate the real-time feasibility of the proposed method, the inference times of the neural SDF, repulsive force calculation, and RL model inference were measured. The neural SDF model efficiently computes the distance between surfaces by leveraging link-index embeddings to calculate SDF values for each link in parallel, achieving an average computation time of 7.17 ms. The repulsive force computation and RL model inference require 0.4487 and 14.21 ms, respectively. Overall, the proposed method achieves real-time performance with a total inference time of 21.38 ms (46.7 fps). The inference was conducted on a computer with an Intel Xeon Gold 6248 CPU (2.50 GHz), 128 GB RAM, and NVIDIA GeForce RTX 3090 GPU.

4 Conclusion

To execute RL-based complex tasks and ensure collision-free motion, this study proposed a novel and effective obstacle representation method based on repulsive forces derived from potential fields. This representation uses a point cloud on obstacle surfaces and a neural SDF to encode the spatial relationship between a robot and the obstacles into a 21-dimensional vector. This approach provides a policy with a fixed-dimensional state representation that remains invariant to the number, shape, and arrangement of obstacles.

The proposed method (85.57%) achieved improved task success rates compared to an RL baseline (73.03%) using PointNet in environments with various obstacles. Moreover, the proposed model led to stable learning for long-horizon manipulation tasks, achieving success rates of 75.3% and 65.0% in two tasks, i.e., opening a door without and with a door lever, respectively, which require sequential motion. These results indicate that the proposed low-dimensional obstacle representation effectively enables both collision avoidance and task execution, demonstrating its scalability to complex industrial environments.

Despite these contributions, several limitations must be addressed. First, this study was conducted under the assumption of a fully observable environment in which the entire point cloud of the obstacles was accessible. The proposed method was not validated under partial observability conditions in which obstacles can be partially occluded or moved outside the field of view of the robot. In such cases, the repulsive force from unobserved obstacle surfaces cannot be considered, potentially degrading the collision avoidance performance. Therefore, future studies should focus on extending this representation to incorporate these uncertainties. Second, this study focused primarily on environments with static obstacles. Although the rapid inference capabilities of RL policies can be easily implemented for slow-moving objects, stable collision avoidance and task execution with fast-moving dynamic obstacles remain challenging. Extending the proposed method to dynamic environments is an important direction for future research.

Despite these limitations, the proposed obstacle-aware representation demonstrates that policies can be effectively trained to perform manipulation tasks without collisions, even in obstacle-rich environments. By compressing high-dimensional geometric relations into a compact repulsive force representation, this method enables stable policy learning while preserving essential spatial relationships between a robot and surrounding obstacles. This characteristic not only improves task success and learning stability, but also highlights the potential of structured geometric representations as an alternative to direct point cloud embeddings for RL-based manipulation. Consequently, the proposed framework provides a promising foundation for scalable obstacle-aware learning in real-world robotic applications, particularly in industrial environments, where safety, generalization, and robustness are critical requirements.

Appendix A: Hyperparameters

This section provides the hyperparameters of the three task policies described in Section 3.5 The reach and opening a door without door lever tasks were trained using PPO, while the opening a door with door lever task was trained using SAC because it requires longer action sequences.

Table 5 Hyperparameters of reach task (PPO)

Hyperparameter	Value
Init noise std	1.0
Value loss coef.	1.0
Entropy coef.	0.003
Clip parameter (ϵ)	0.2
Gamma (γ)	0.99
Lambda (λ)	0.95
Learning epochs	5
Mini-batches	4
Max gradient norm	1.0
Target KL	0.01
Learning rate	3.0e-4
Actor dimension	(64, 64)
Critic dimension	(128, 64)

Table 6 Hyperparameters of Opening a door without door lever (PPO)

Hyperparameter	Value
Init noise std	1.0
Value loss coef.	1.0
Entropy coef.	0.001
Clip parameter (ϵ)	0.15
Gamma (γ)	0.99
Lambda (λ)	0.95
Learning epochs	5
Mini-batches	4
Max gradient norm	1.0
Target KL	0.008
Learning rate	1.0e-4
Actor dimension	(128, 128)
Critic dimension	(256, 256)

Table 7 Hyperparameters of Opening a door without door lever (PPO)

Hyperparameter	Value
Gamma (γ)	0.99
Tau (τ)	0.005
Learning epochs	5
Batch size	128
Gradient step	1
Buffer size	1.0 e+6
Learning starts	3300
Learning rate	1.0e-4
Actor dimension	(256, 256)
Critic dimension	(256, 256)

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Data Availability The data will be available on reasonable request.

Declarations

Competing interest The authors declare that they have no competing financial interests or personal relationships.

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